Georgia Institute of Technology

School of Computer Science

CS 4290/6290, ECE 4100/6100: Spring 2021 (Prof. Conte)

Practice

1. Answer the following questions:
   a) In what situations is directory-based coherence better than snoop-based coherence? Why?
   b) In a system with 1TB of memory, 256 cores, and a cache line size of 32 bytes, how many state bits does a directory protocol (MOESI) impose on each cache line? What is the overhead ratio of state bits to data bits?
   c) (Continue from part b) To reduce overhead, consider a limited pointer scheme that supports 8 cores per cache line. How many bits are required to track 256 cores for each cache line? What is the overhead ratio of state bits to data bits?
   d) (Continue from part c) What happens when the number of shares on a cache line exceeds 8? What are some ways to resolve this?

2. Calculate the number of cycles required for a SI write through, write update cache coherence protocol on the following memory reference streams:

   a) r1 r2 r3 w1 w2 w3 r1 r2 r3
   b) r1 w1 r2 w2 r3 w3 r4 w4

   Each processor uses a write-through cache. All the references are to the same location: r/w indicates read or write and the digit refers to the processor issuing the reference. Assume that all caches are initially empty and use the following cost model: cache hit: 1 cycle, transfer of full cache block across bus: 70 cycles, all other bus transactions (write-through, write update): 50 cycles. Although it does not matter in the question, you can assume that the protocol allows cache to cache transfers. Assume that write through and write update operations together take 50 cycles.

3. In this question you should describe the effects of different cache coherence protocols. All accesses are made to the same block. RX means a read executed by processor X, and WX means a write executed by processor X.

   The table below describes the access trace; you must select one of the options on each cell (or just include a table with your answer for each cell). The cells reflect the states/actions that were caused by the instruction.

a) Assume you have 3 processors with a write invalidate **MSI** protocol and write-back private caches.

Instr	P1	P2	P3	Read from mem?	Write to mem?
R3	M/E/S/I	M/E/S/I	M/E/S/I	Yes/No	Yes/No
W3	M/E/S/I	M/E/S/I	M/E/S/I	Yes/No	Yes/No
R2	M/E/S/I	M/E/S/I	M/E/S/I	Yes/No	Yes/No
W2	M/E/S/I	M/E/S/I	M/E/S/I	Yes/No	Yes/No
W3	M/E/S/I	M/E/S/I	M/E/S/I	Yes/No	Yes/No
W1	M/E/S/I	M/E/S/I	M/E/S/I	Yes/No	Yes/No
R2	M/E/S/I	M/E/S/I	M/E/S/I	Yes/No	Yes/No

b) Now, assume you have 3 processors with a write invalidate **MESI** protocol and write- back private caches.

Instr	P1	P2	P3	Read from mem?	Write to mem?
R3	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
W3	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
R2	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
W2	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
W3	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
W1	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
R2	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No

c) Now, assume you have 3 processors with a write invalidate **MOESI** protocol and write-back private caches (AMD Opteron uses this protocol).

Instr	P1	P2	P3	Read from mem?	Write to mem?
R3	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
W3	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
R2	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
W2	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
W3	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
W1	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No
R2	M/O/E/S/I	M/O/E/S/I	M/O/E/S/I	Yes/No	Yes/No

4. Using Load Link, Store Conditional, implement the function FetchAndAdd(x, d), which automicly adds d to x and stores the result in x. In class, we said that LL, SC work well with a snooping coherence protocol, but need special messages for a directory-based protocol. Why?